SPRINT 8 TEAM 1A

Sprint 8

Team 1 Practical A

Project Update and Team Progress

R2 — SCRUM MASTER

Sprint 8: Goal & Delivery Snapshot

The sprint's goal was consolidation — improve and merge existing work rather than add new scope, and ship a clean, documented W30 release.

8+

PRs Merged

Database storage, macro fix, human score, archive weeks, Delta export, artifacts

7

Daily Standups

Tracked every day — blockers surfaced and logged

8

Wiki Pages

New GitHub Wiki created; README simplified

L4

Intelligent System

Live dashboard + Human Score integrated into output

*After a difficult Sprint 7, the team delivered consistently this sprint — **work merged steadily rather than piling up at the deadline.***

R2 — SCRUM MASTER

Retrospective & Process Change

WHAT WENT WELL

- Steady progress throughout the sprint
- Core features improved and integrated
- Strong collaboration across roles

PROBLEMS WE FACED

- Some PRs needed extra review rounds
- Limited free model availability
- Unclear task ownership delayed one item to the deadline

LEARNINGS & ACTIONS

- Confirm task ownership when assigning
- Keep documentation current

PROCESS CHANGE FOR SPRINT 9

Free LLM models on OpenRouter ran low this sprint, so the team identified backup models and will keep alternatives configured to prevent pipeline stalls. Blockers are now raised in the main channel the day they occur, not at standup.

R1 — PRODUCT OWNER

Product Owner Update · R1 Sprint Goal: System Hardening

This sprint is centered on a single, high-impact objective: hardening the end-to-end Market Intelligence pipeline around a **single source of truth** and finalizing the prediction range model to deliver reproducible, auditable weekly predictions at scale.

Database Integration

All local runtime state is being migrated entirely into an **SQLite database**. Every non-image artifact — including LLM raw outputs, Final Predictions, and Human Scores — is now both persisted in and re-exported from the DB.

Agent Upgrades: Prediction Range

The full prediction range schema — covering **direction, percentage range, and confidence level** — is now implemented end-to-end across all three analytical agents: **Almanac**, **Technical**, and **Macro**.

System Reliability: Run Restoration

Reopening a past or interrupted pipeline run now correctly restores **true stage progress** from the database, replacing the prior behavior where all stages incorrectly reported as complete.

Acceptance Criteria: Definition of Done

The vW30 sprint closes when both the **"New" criteria** (structural pipeline upgrades) and the **"Usual" criteria**(weekly output completeness) are fully satisfied. All criteria below must be verified and committed before the release tag is applied.

The "New" Criteria — Structural Upgrades

These acceptance criteria represent the new architectural baseline introduced this sprint. They must be met for the sprint goal to be considered achieved.

- **SQLite artifact handling:** The Agent base class manages all artifact parsing and persistence via SQLite — no direct file I/O for structured data outside of the DB layer.
- **All pipeline artifacts export:** All 4 agents, llms, human score, delta and final prediction are exported successfully

The "Usual" Criteria — Weekly Output Completeness

These are the standing quality gates applied every sprint to ensure the weekly release package is complete, auditable, and ready for stakeholder consumption.

- **Sprint documentation:** Sprint goal and acceptance criteria documented cleanly and committed.
- **Prediction coverage:** Direction, range, and confidence locked for **SPX**, **NDX**, and **IWM**.
- **LLM evidence:** All 4 raw LLM responses and associated chart evidence safely stored in the DB.
- **Human Score:** Completed with a documented override justification where applicable.
- **Release tag:** System sealed with the `vW30` release tag before the sprint deadline.

Release Gate: The `vW30` tag must not be applied until every criterion above is verified. Partial completion does not constitute a closed sprint.

R1 — Product Owner

What We Delivered This Sprint

The sprint's work grouped by area — everything below was completed and merged into main.

✓	**Backend & Data**	Local database storage · persistence · archive-week handling · artifacts extraction
✓	**UI & UX**	Final prediction · Horizon Selector · Multiple runs · Archive load fixes
✓	**Documentation**	New 8-page GitHub Wiki · simplified, reorganised README · usage & setup guides

*Work merged steadily into main throughout the sprint — **kept current through daily standups.***

Market Intelligence Sprint 8

WEEK 30 UPDATE

A multi-agent synthesis covering seasonal context, macroeconomic environment, technical trend analysis, W30 actuals, LLM consensus regime, and calibration delta — designed to inform tactical positioning for institutional traders and market analysts heading into a high-volatility week.

📅 Cycle

Midterm Election Year — Q2–Q3 Weak Spot

📡 Regime

Bearish Consensus — Medium Confidence

⚡ Risk Events

Fed FOMC July 29 + Mega-Cap Earnings

📊 Trend

Zone 4 Bearish — All Major Indices

R3 — ALMANAC AGENT

Seasonal Context: Mixed Signals in a Midterm Year

The Almanac Agent places current market conditions within a historical seasonal framework. July of a **midterm election year** lands squarely inside the **Q2–Q3 "Weak Spot"** — a recurring pattern in which broad equity performance tends to underperform relative to other seasonal windows. This structural headwind is a persistent background input that, by itself, does not drive trade decisions but helps contextualize the risk/reward environment.

Sector-level seasonal signals are notably divergent this week, which is the primary driver of the agent's **Mixed Almanac Bias** and **Low Confidence** rating. With macro events of the magnitude of a Fed decision and a mega-cap earnings cluster on deck, seasonal patterns are highly susceptible to overrides. Traders should treat almanac signals as directional color rather than actionable triggers in isolation this sprint.

⚠ Seasonal signals carry **Low Confidence** this week. Macro catalysts — particularly the July 29 FOMC and tech earnings — are the dominant regime drivers and can easily override historical patterns.

📅 Cycle Context

Midterm election year. July currently sits inside the **Q2–Q3 Weak Spot** — historically a period of suppressed broad-market returns.

🟢 Seasonal LONG

Technology (XLK) — Bullish seasonal window active. This is the sole sector with a constructive seasonal tailwind entering W30.

🔴 Seasonal SHORT

Financials, Materials, Energy, Gold — All four sit within bearish seasonal windows, reinforcing caution across cyclical and commodity-linked exposures.

⚖️ Almanac Bias

Mixed. Conflicting sector signals with a weak broad-market cycle backdrop. Low confidence; susceptible to macro overrides this week.

R4 — MACRO AGENT

Economic Environment: Fed Decision Dominates

The macro landscape heading into W30 is defined by a single overriding catalyst: the **July 29 FOMC meeting**. With a 66% probability of a rate hold priced in, markets are not necessarily expecting a surprise — but the statement, tone, and any shifts in the dot plot will carry outsized weight given the elevated 10-year yield environment. The macro agent assigns a **Neutral Bias** with **Medium Confidence**, acknowledging that the primary directional driver is event-based rather than trend-based.

🏦 Fed & Rates

Next FOMC: **July 29**. Market pricing reflects a **66% probability of a Rate Hold for 3.50%-3.75%**, down from **87%** the previous week, indicating increased expectations of a hike to **3.75%–4.00%**. The 10-year Treasury yield is rising, last observed at **4.71%** — a level that continues to exert valuation pressure on growth and long-duration equities.

🛢️ Commodities

WTI Crude Oil surged a dramatic **+8.27%** in the prior week — a significant macro shift that benefits Energy but adds cost-push inflation risk. The Dollar (DXY) also firmed, rising **+0.69%**, creating a modest headwind for international earnings and emerging market assets.

💻 Earnings Risk

A **massive mega-cap tech earnings week** adds a second major volatility vector. **MSFT, META, AAPL, and AMZN** all report during W30, concentrating idiosyncratic event risk in the largest-weight index constituents and elevating implied volatility for NDX in particular.

📊 Macro Bias

Neutral — Medium Confidence. The macro environment is not directionally decisive on its own. The primary driver is the binary outcome of the Fed Interest Rate Decision, which could rapidly shift the bias in either direction.

R5 — TECHNICAL AGENT

Trend Overview: Zone 4 Bearish Across All Indices

The Technical Agent delivers the most unambiguous signal of this sprint: **Zone 4 (Bearish) conditions across all three major indices** — SPX, NDX, and IWM. This is not a mixed or borderline reading. All indices are trading below their short-term moving averages, and the 8-day EMA has crossed below the 21-day EMA for each — a classic momentum confirmation signal that institutional trend-following systems treat as a sell or avoid signal.

The distinction in confidence levels is worth noting: **NDX carries the highest confidence (High)**, reflecting the concentrated weight of mega-cap tech and the severity of its recent breakdown relative to its EMA stack. **SPX and IWM are Medium confidence**, acknowledging that both remain closer to support zones where a reversal is possible — particularly if the Fed delivers a dovish surprise or earnings beats trigger a sentiment shift. Nevertheless, the burden of proof currently sits with the bulls.

The 8 EMA trading below the 21 EMA across all three indices simultaneously is a high-conviction bearish momentum confirmation. This pattern historically precedes continued downside unless broken by a strong catalyst-driven reversal.

📉 Trend Condition

Zone 4 — Bearish across SPX, NDX, and IWM. No index is in a constructive trend posture entering W30.

📊 Moving Averages

SPX, NDX, and IWM are all trading **BELOW their 8-day and 21-day EMAs**. The 8 EMA has crossed below the 21 EMA on all three — confirming downward momentum is not a short-term spike but a structural breakdown.

🔴 Technical Bias

Bearish. No ambiguity at the index level. The path of least resistance is lower absent a catalyst-driven recapture of key EMAs.

🎯 Confidence Levels

SPX: Medium. **IWM:** Medium. **NDX:** High. NDX exhibits the cleanest and most decisive breakdown, making it the highest-confidence bearish read of the three.

R5 — TECHNICAL AGENT

Key Levels: Critical Support, Resistance & Invalidation

The following table presents the precise price levels that define the technical landscape for W30. **Resistance 1** marks the level where a rally is expected to stall or reverse. **Support 1** is the nearest downside level to defend. The **Invalidation Level** — a daily close above the Resistance 1 level — is the signal that would neutralize the current bearish bias and require a full reassessment of the trend regime.

Index	Last Close	Resistance 1	Support 1	Invalidation Level
SPX	7,408	7,582	7,294	Daily Close > 7,582
NDX	28,455	30,329	28,231	Daily Close > 30,329
IWM	292.09	302.72	290.17	Daily Close > 302.72

Distance to Support: All three indices are trading close to their nearest support levels — SPX has roughly 114 points of downside to Support 1, NDX has 224 points, and IWM has approximately $1.92. This compressed downside buffer is notable: a single adverse macro catalyst (e.g., a hawkish Fed surprise) could test support levels within the same session.

Distance to Invalidation: Recovery rallies face substantial overhead resistance. SPX would need to recapture 174 points, NDX 1,874 points, and IWM $10.63 to invalidate the bearish trend regime. These are not trivial hurdles under current EMA conditions, reinforcing the asymmetric risk skew to the downside for W30.

R6 — DATA AGENT

W30 Actuals: Broad Weakness With Pockets of Strength

-0.66%

SPX Return

S&P 500 weekly performance, W30

-0.48%

NDX Return

Nasdaq 100 weekly performance, W30

+8.27%

WTI Surge

Crude Oil weekly move — dominant commodity driver

5/11

Green Sectors

Mixed breadth — exactly half the market finished positive

Sector Extremes

The sector performance landscape in W30 was sharply bifurcated. **Energy (+2.95%)** led all sectors, directly benefiting from the WTI crude surge. **Utilities (+2.26%)** continued their defensive outperformance, suggesting rotation into safety-oriented income plays. **Technology (+1.63%)** closed green, providing a notable contrast to the bearish technical posture of NDX — consistent with the almanac's bullish seasonal window for XLK. On the other end, **Consumer Discretionary (-5.79%)** suffered the largest weekly loss, a signal of deteriorating consumer sentiment and earnings-driven pressure. **Communication Services (-4.76%)** was the second-worst performer, reflecting pre-earnings anxiety around mega-cap tech names.

Market Drivers & Breadth

Bonds (TLT) dropped -1.60%, consistent with the rising 10-year yield narrative and a continued repricing of rate expectations. **VIX cooled slightly (-1.01%)**, but remains elevated enough to signal investor caution heading into the FOMC and earnings cluster. The fact that exactly **5 out of 11 sectors finished green** reflects a genuinely mixed breadth environment — not a broad risk-off rout, but also not a healthy risk-on rally. Selective rotational activity is underway, with money moving from growth and discretionary into energy and defensive yield plays.

R8 — LLM SYNTHESIS

Consensus Regime: Bearish — Medium Confidence

Four independent large language models were queried as part of the R8 synthesis engine: **Nemotron 3 Super, Ling 3.0 Flash, Gemma 4, and Laguna M.1.** Each model independently processed the available macro, technical, seasonal, and data agent outputs to generate a probabilistic weekly regime call and forward range estimate for SPX, NDX, and IWM.

The models converged on a **Bearish consensus with Medium Confidence** — a regime call that aligns with and reinforces the Technical Agent's Zone 4 reading. The core narrative across models is consistent: *bearish technical patterns outweigh the mixed seasonal cues, and the high-volatility event risk from the July 29 Fed decision and mega-cap earnings cluster creates a downside-skewed risk/reward for the week.*

The **key contradiction** that all models flagged: Technology (XLK/NDX) carries a **bullish seasonal tailwind** from the Almanac Agent, yet the NDX is simultaneously the **most technically bearish** of the three indices with the highest-confidence breakdown reading. This divergence creates an unstable setup where a strong earnings beat from MSFT, META, AAPL, or AMZN could trigger a rapid directional reversal — making position sizing and stop discipline critical for W30.

Models Queried

- Nemotron 3 Super
- Ling 3.0 Flash
- Gemma 4
- Laguna M.1

Consensus Weekly Regime

▼ BEARISH

Medium Confidence

Key Contradiction

Tech has **bullish seasonality** (XLK Almanac LONG) but NDX is **technically bearish** (Zone 4, High Confidence). This divergence is the single greatest source of regime uncertainty for W30.

R8 — LLM SYNTHESIS

Model Comparison: Forward Range Estimates by Asset

Each model produced independent forward range estimates for SPX, NDX, and IWM, along with a primary catalyst focus. These ranges represent the models' probabilistic weekly outcomes — not point predictions — and should be read as a distribution of outcomes rather than a single forecast. Wider ranges indicate higher model uncertainty.

Model	SPX Range	NDX Range	IWM Range	Key Catalyst Focus
Nemotron 3 Super	-0.8% to +0.4%	-0.3% to +0.5%	-0.8% to +0.2%	EMA Breakdowns
Ling 3.0 Flash	-1.5% to +0.5%	-2.5% to +0.5%	-1.5% to +0.5%	Fed Event Risk
Gemma 4	-2.5% to +0.5%	-4.0% to 0.0%	-2.0% to +0.8%	Tech Earnings Volatility
Laguna M.1	-2.0% to +0.5%	-1.5% to +0.8%	-2.5% to +0.3%	Energy/Tech Divergence

Downside Skew: All four models assign asymmetric downside in their ranges. The bearish scenarios are consistently larger in magnitude than the bullish scenarios — ranging from a tight -0.8% (Nemotron, base case) to a severe -4.0% for NDX (Gemma). This skew is mathematically consistent with the Zone 4 technical reading and the event-driven volatility premium embedded in options pricing heading into FOMC week.

Model Divergence: The widest disagreement is in the NDX range: Nemotron sees a relatively contained -0.3% to +0.5% move, while Gemma models a potential -4.0% to 0.0% outcome — a 350 basis point spread. This divergence reflects genuine model uncertainty about the magnitude of the tech earnings reaction. Laguna's focus on **Energy/Tech Divergence** is the most novel catalyst thesis, potentially identifying a cross-sector rotation trade for the week.

R7 — HUMAN SCORE

Human Score — Week 30

AI CONSENSUS	HUMAN CALL	HUMAN SCORE TOTAL
Bearish	**Bearish · Medium**	**+1**

FIVE-DIMENSION JUDGEMENT

Macro / News Weight	AI: Neutral	0
Technical Structure	AI: Bearish	0
Almanac Seasonal Weight	AI: Mixed	-1
AI Model Agreement	AI: 4 of 4 Bearish	+2
Wild Card / Observation	AI: none flagged	0

Total +1 — *the team agreed with the AI's Bearish call, driven by confirmed technical structure and unanimous 4-model agreement.*

R7 — HUMAN SCORE

Human Judgement & Walkthrough

OVERRIDE

Team agrees with the AI consensus (Bearish), driven by confirmed bearish technical structure and unanimous agreement across all four models. Confidence held at Medium — not High — due to mixed macro signals and low-confidence Almanac data.

INVALIDATION CONDITION

The bearish prediction would be wrong if the S&P 500 closes above the resistance level of about **7,581**, or if the Federal Reserve gives more positive news than expected on **July 29**. All four AI models said the Fed decision is the main event that could change the market direction.

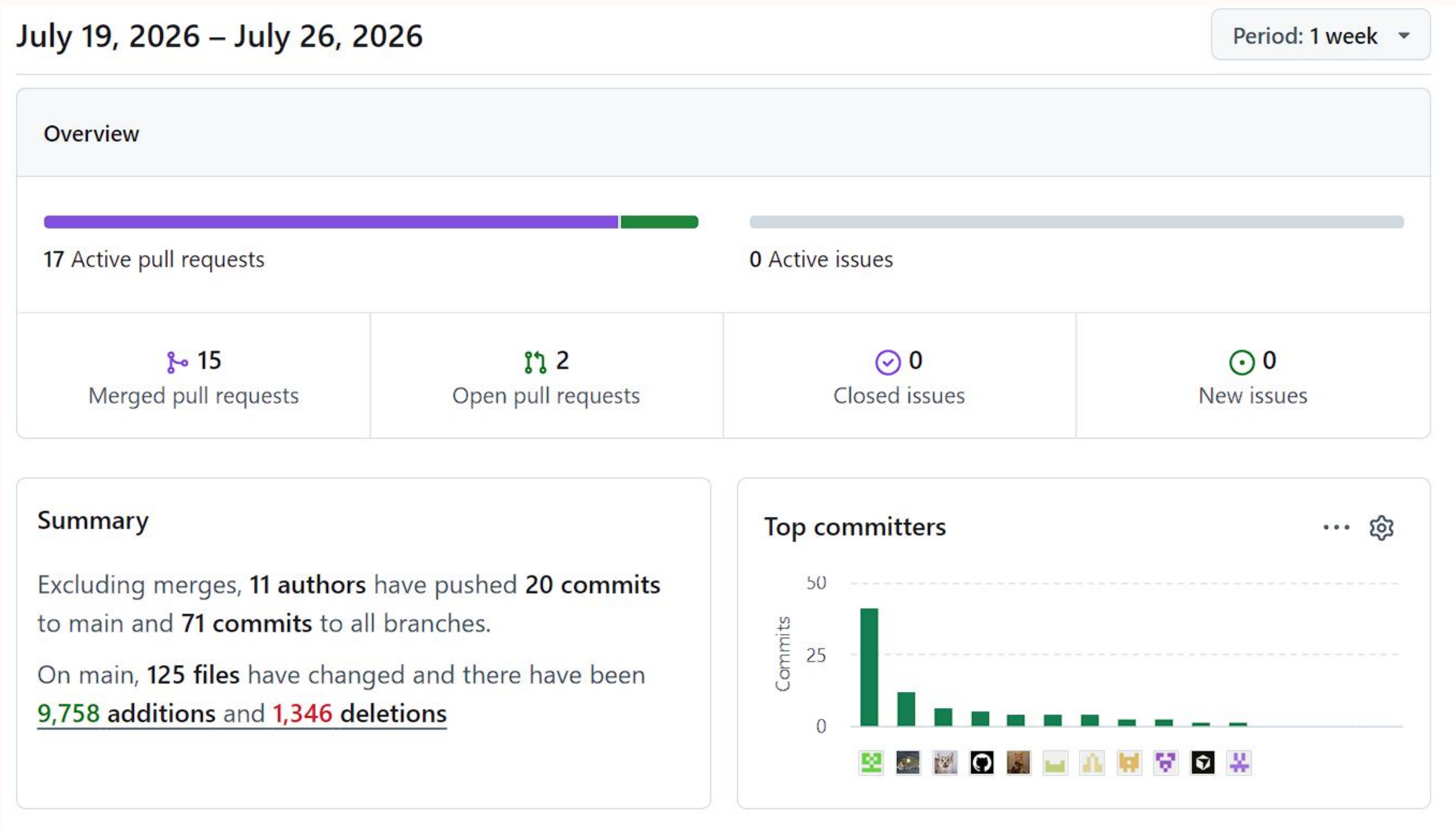
HUMAN SCORE WALKTHROUGH

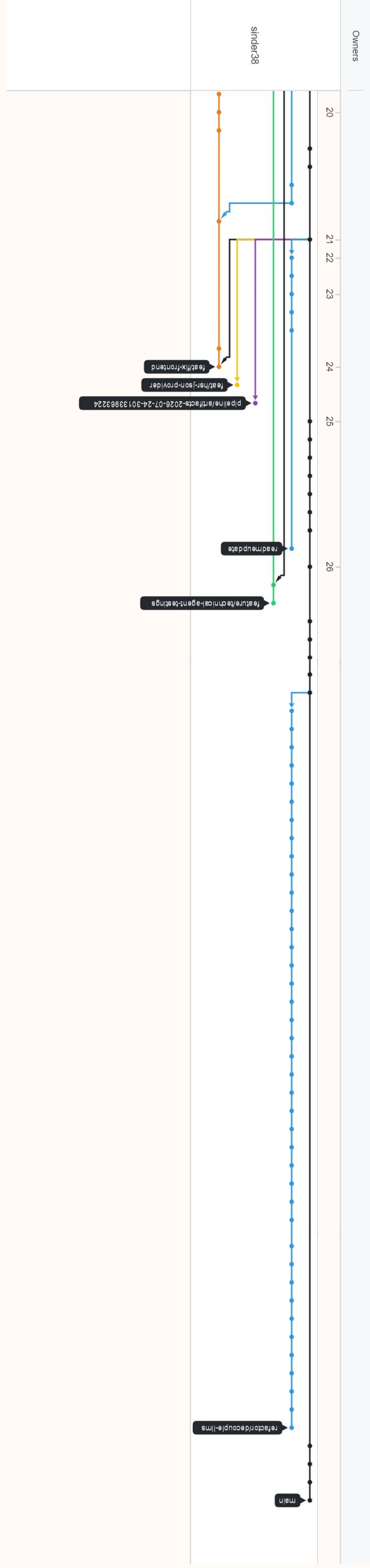
60-second walkthrough of the Week 30 Human Score.

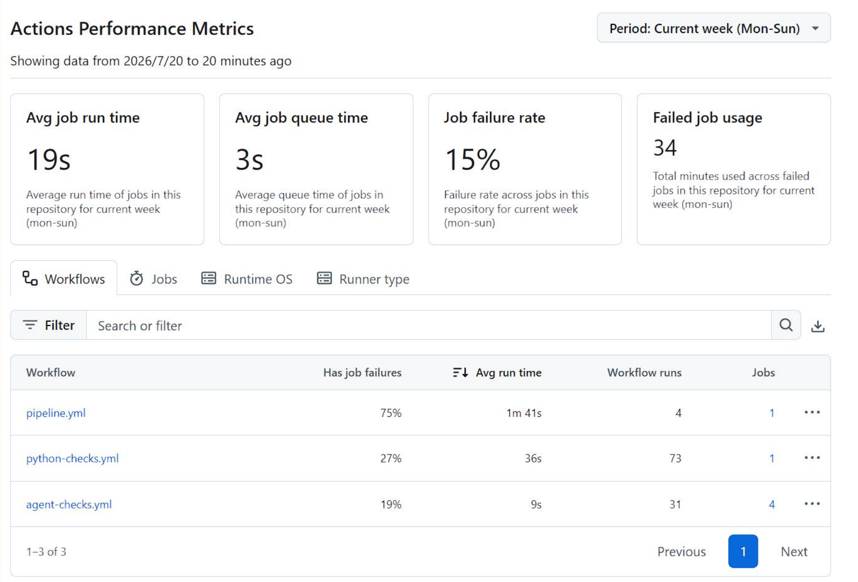
R9 — GITHUB LEAD

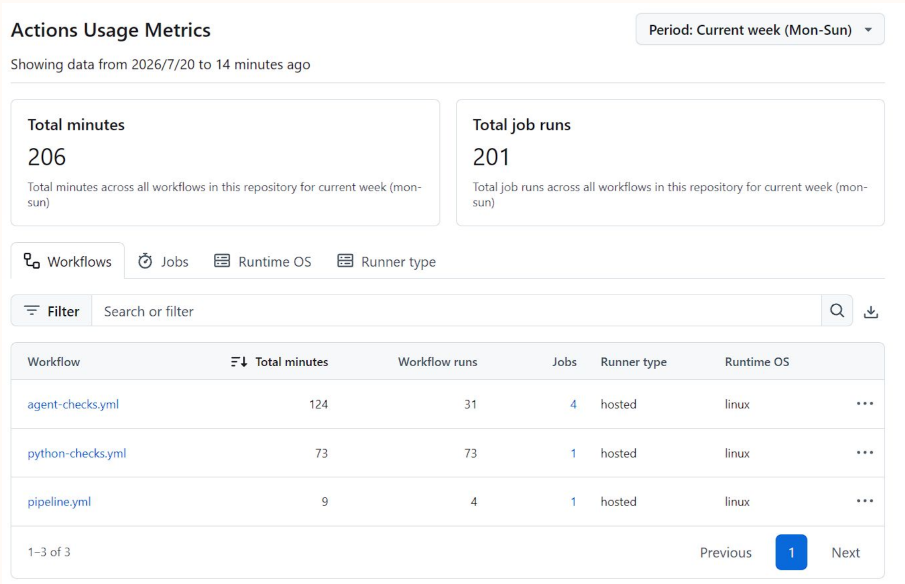
GitHub Repository

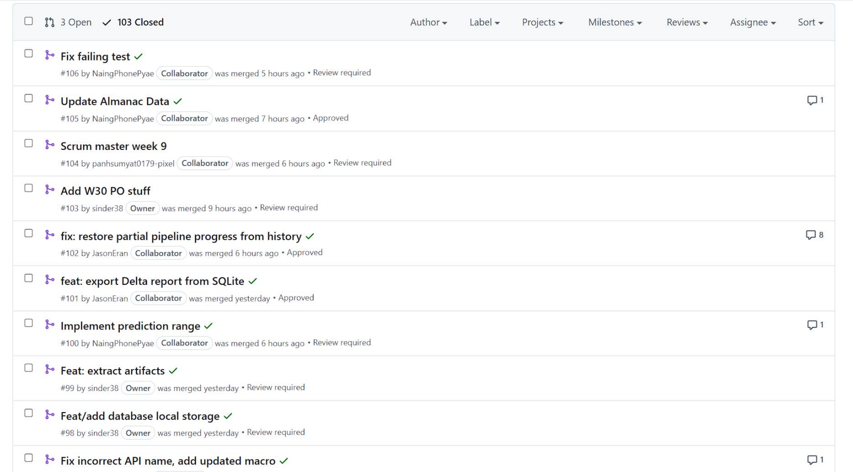
- **System Database Integration:** All non-image local data is stored in the SQLite database, and parsing is part of the Agent base class
- **Refined System Accuracy:** Opening a past or interrupted run from history restores its real stage progress and does not show false completion
- **AI Provider Correction:** Updated 4 LLM models & Ollama.
- **Improvement on Macro Agent:** API retrieval errors will no longer occur.
- **Artifact Export Availability:** Accessibility to extract artifacts out of the database into the data/ dir missing delta, final prediction, human score.

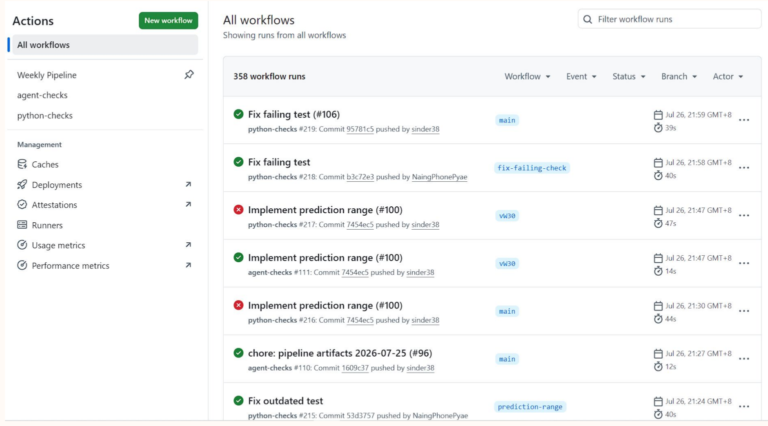










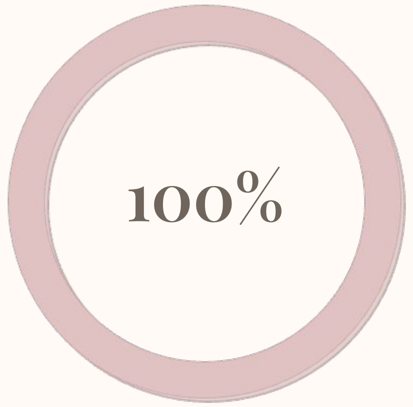


[GitHub Repo Link]()

R10 — CALIBRATION AGENT

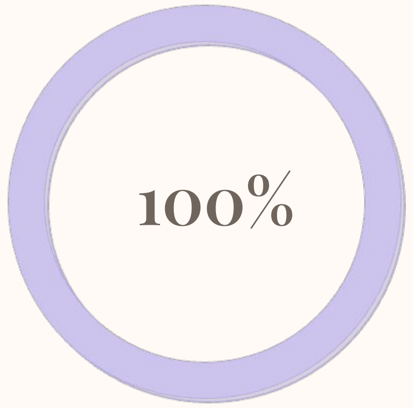
Delta Engine: W29 Performance & Weight Recalibration

W29 Prediction Accuracy



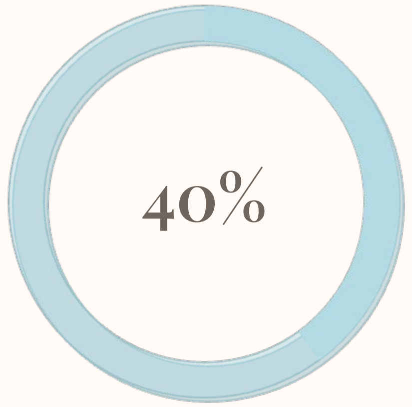
Direction Accuracy

3/3 assets correct — SPX, NDX, IWM



Range Accuracy

3/3 assets within predicted range



Cumulative Accuracy

Direction & Range combined — rebounding trend

Agent Weight Recalibration — Sprint 8 Adjustments

Following the W29 perfect-week accuracy result, the Delta Engine has recalibrated agent weights for Sprint 8. The adjustments reflect the relative signal quality and predictive contribution of each agent over the trailing calibration window.

Agent	Previous Weight	New Weight	Direction
Technical Agent	0.30	**0.35**	▲ +0.05
Human Score	0.20	**0.25**	▲ +0.05
Almanac Agent	0.15	**0.10**	▼ -0.05
LLM Synthesis	0.15	**0.10**	▼ -0.05

The increase in **Technical Agent (+0.05)** and **Human Score (+0.05)** weights reflects their stronger recent predictive performance — particularly the Technical Agent's clean Zone 4 calls. The reduction in **Almanac (-0.05)** and **LLM (-0.05)** weights acknowledges that both delivered noisier, lower-confidence signals during the calibration window. These adjustments are incremental and will be re-evaluated after Sprint 9 actuals are recorded.

SPRINT 8

TEAM 1A

Thank You

Any Questions?